

JavaScript — Practice Assignments

45 Days Internship For Web Development

5 Unique Mini-Project Assignments • Attempt Any 3

Assignment 1: Lost Treasure Expedition | Assignment 2: Movie Studio Management

Assignment 3: Smart Supermarket Billing | Assignment 4: Mars Rover Navigation

Assignment 5: Internship Recruitment Portal

⚠ **Attempt only 3 out of 5 assignments — Submit before 17th July 2026, 11:59 PM**

45 Days Internship For Web Development — Assignments

Instructions

READ FIRST

⚠ **Important:** Attempt **only 3 out of the 5** assignments given below (your choice).
Deadline: Submit before **17th July 2026, 11:59 PM**. Late submissions will not be accepted.

Ye 5 mini-project assignments hain jisme aapko is course mein seekhe gaye saare concepts — forms, arrays, objects, DOM, timers, dates, aur error handling — ek real project ki tarah use karne hain.

Assignments Overview

#	Assignment	Core Theme
1	The Lost Treasure Expedition	Random game logic + timers
2	Movie Studio Management System	Booking system + pricing logic
3	Smart Supermarket Billing System	Billing, GST & discount calculation
4	Mars Rover Navigation Simulator	State tracking + live updates
5	Internship Recruitment Portal	CRUD operations + ES6 features

Tip: Assignment choose karte waqt wahi 3 pick karo jisme aapko concepts sabse zyada practice karne hain — sab assignments equally weighted hain.

45 Days Internship For Web Development — Assignments

Assignment 1: The Lost Treasure Expedition

OPTIONAL — PICK 3

Ek treasure-hunting game banao jisme user apni details register kare, random treasure code guess kare, tools manage kare, aur expedition ka time track kare.

Questions

1. Registration form banao (Name, Age, Country). Agar Age < 16 ho toh "Not Eligible" dikhao.
2. Random treasure code generate karo (1-50 ke beech). User ko 6 attempts do — har guess pe "Too High" / "Too Low" / "Found!" dikhao.
3. Tools ka array banao — Add, Remove, Search, aur Display functions likho.
4. Explorer object banao: Name, Age, Points, Level, Tools.
5. 60-second countdown timer banao (setInterval / setTimeout se).
6. Current date aur expedition ka start/end time show karo.
7. Invalid inputs ko try...catch se handle karo (jaise empty name, negative age).

Concepts Used

- Forms & Input validation, Arrays & Array methods
- Objects, setInterval/setTimeout (Timers)
- Date object, try...catch Error Handling

45 Days Internship For Web Development — Assignments

Assignment 2: Movie Studio Management System

OPTIONAL — PICK 3

Ek movie ticket booking system banao jisme movies manage ho, tickets book ho discount ke saath, aur booking details DOM par dikhayi jaye.

Questions

1. Booking form banao (Name, Movie, Age, Tickets).
2. Movie array banao — Add, Delete, Search, Display functions likho.
3. Ticket price calculate karo discounts ke saath (jaise senior citizen / bulk booking discount).
4. Booking object banao jisme sabhi booking details ek saath store ho.
5. Booking ko DOM manipulation se page par show karo.
6. 2-minute booking timer banao (jitne time mein booking confirm karni ho).
7. Invalid data (jaise negative tickets, empty movie name) ke liye custom errors throw karo.

Concepts Used

- Forms & Input validation, Arrays & Array methods, Objects
- DOM Manipulation, Timers, throw / try...catch

45 Days Internship For Web Development — Assignments

Assignment 3: Smart Supermarket Billing System

OPTIONAL — PICK 3

Ek supermarket billing system banao jisme grocery items manage ho, bill calculate ho GST/discount ke saath, aur final bill DOM par generate ho.

Questions

1. Grocery array banao prices ke saath.
2. Products ko Add, Remove, Update, Search karne ke functions likho.
3. Total, GST, Discount, aur Final Amount calculate karo.
4. Customer object banao (name, items, total, etc.).
5. String methods apply karo (jaise product name trim/format karna).
6. 90-second offer timer banao (limited-time discount ke liye).
7. DOM manipulation se final bill generate karke dikhao.
8. Custom errors handle karo (jaise invalid quantity, product not found).

Concepts Used

- Arrays & Array methods, Arithmetic Operators, String methods
- Objects, DOM Manipulation, Timers, Error Handling

45 Days Internship For Web Development — Assignments

Assignment 4: Mars Rover Navigation Simulator

OPTIONAL — PICK 3

Ek Mars Rover simulator banao jisme rover ko control buttons se move karaya jaaye, battery drain ho, aur live coordinates track ho.

Questions

1. Rover object banao (coordinates, battery, direction, etc.).
2. Control buttons banao (Forward, Backward, Left, Right).
3. Movement history ko array mein store karo.
4. Har move ke baad battery ko 5% se reduce karo.
5. Live coordinates page par show karo.
6. Har 5 second mein battery auto-drain karo (setInterval se).
7. Mission ka start/end time show karo.
8. Invalid commands (jaise battery 0% pe move) ke liye error throw karo.

Concepts Used

- Objects, Event Handling (buttons), Arrays
- Timers, Date object, throw / try...catch

45 Days Internship For Web Development — Assignments

Assignment 5: Internship Recruitment Portal

OPTIONAL — PICK 3

Ek recruitment portal banao jisme candidates register ho, eligibility check ho, aur interview dates ka countdown dikhaya jaaye.

Questions

1. Candidate registration form banao.
2. Candidates ko array of objects mein store karo.
3. Eligibility check karo (jaise minimum age/qualification).
4. Add, Update, Delete, Search functions likho.
5. Candidates ko DOM manipulation se display karo.
6. ES6 features use karo (arrow functions, destructuring, spread operator, template literals).
7. Registration/interview dates aur countdown show karo.
8. Duplicate IDs aur invalid inputs ko handle karo (try...catch / throw).

Concepts Used

- Arrays of Objects, DOM Manipulation, ES6 Features
- Date object, Error Handling (duplicate/invalid checks)

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Submission Guidelines

FINAL STEP

⚠ **Reminder:** Attempt **only 3 out of the 5** assignments. **Deadline: 17th July 2026, 11:59 PM.**

Project Structure

Har assignment ke liye HTML, CSS, aur external JS use karo, proper comments likho, DOM manipulation implement karo, aur errors ko `try...catch` se handle karo.

File / Folder	Purpose
index.html	Page structure
style.css	Styling
script.js	External JavaScript logic
assets/	Images, icons ya other media files
README.md	Project description aur instructions

Har assignment ke liye alag folder banao (jaise `assignment-1/`, `assignment-2/`) aur upar diye gaye 5 files/folders har ek ke andar submit karo.

Quick Checklist

- Sirf 3 assignments attempt kiye (5 mein se)
- HTML + CSS + External JS use kiya
- Proper comments aur DOM manipulation kiya
- `try...catch` se errors handle kiye
- Deadline se pehle submit kiya — 17th July 2026, 11:59 PM